



*In
association with*



Presents

AI /ML and Design Thinking

Aryabhata Young Innovators Challenge (AYIC)

(Phase Wise Execution Road map)

1. Executive Summary

The National AI / ML School Innovation Challenge is proposed as a structured, pan-India student innovation initiative designed to develop practical skills in Artificial Intelligence, Machine Learning and Design Thinking. The students will engage 800 eligible schools across four geographic regions of India, with one student and one project representing each school.

The challenge will move participants through a defined journey of problem discovery, human-centered design, AI/ML solution development, prototyping, evaluation and national-level recognition. The objective is to create a credible platform where students can convert real-world problems into technology-enabled solutions.

Event Vision

To establish a national platform that empowers school students to become responsible innovators by applying AI/Machine Learning and Design Thinking to meaningful real-world challenges.

Key Objectives

- Promote practical understanding of AI and Machine Learning among school students.
- Develop creativity, critical thinking, problem-solving and innovation capabilities.
- Introduce students to human-centered Design Thinking and structured project development.
- Encourage responsible and impact-oriented use of emerging technologies.
- Create a national platform for schools, educators, industry experts and innovation partners.
- Identify and recognize high-potential student innovators and scalable ideas.

National Participation Framework

India Region	Eligible Schools	Student Representatives	Projects
North India	First 200	200	200
South India	First 200	200	200
East India	First 200	200	200
West India	First 200	200	200
TOTAL	First 800	800	800

Participation principle: One School → One Registration → One Student → One Project.

2. Phase-Wise Event Road map

PHASE 01 | STRATEGY, PLANNING & GOVERNANCE

- Finalize event theme, objectives, eligibility and competition rules.
- Define four regional zones and establish a 200-school capacity per region.
- Develop project handbook, submission requirements, jury framework and evaluation rubric.
- Finalize awards, certificates, branding, partner requirements and communication assets.
- Set up the official website/registration and project management workflow.

PHASE 02 | NATIONAL SCHOOL OUTREACH & CAMPAIGN

- Launch the national communication campaign across four regions.
- Start advertisement on social media platform like Instagram, Facebook, LinkedIn and Meta.
- Engage school principals, coordinators and institutional partners.
- Distribute official invitation, brochure, project guidelines and event calendar.
- Track school outreach and registration capacity region-wise.

PHASE 03 | REGISTRATION & ONBOARDING

- Open official school registration.
- Collect school, coordinator, student and initial project information.
- Enforce one-student/one-project participation per school.
- Generate a unique registration ID for each participating school.
- Share confirmation and project development guidelines.

PHASE 04 | STUDENT ORIENTATION & CAPACITY BUILDING

- Conduct a national launch/orientation session.
- Introduce AI/ML, responsible innovation and Design Thinking.
- Explain project standards, judging process and submission expectations.
- Provide expert-led workshops on problem identification, ideation and prototyping.

PHASE 05 | PROJECT DEVELOPMENT & MENTORING

- Identify and research a genuine real-world problem.
- Apply the Design Thinking process from empathy through testing.
- Develop an AI/ML-enabled concept, prototype or working solution.
- Document data, methodology, model/application, testing and impact.
- Conduct progress reviews and mentor checkpoints.

PHASE 06 | SUBMISSION, COMPLIANCE & SCREENING

- Collect final project submissions through the official platform.
- Verify eligibility, documentation and technical compliance.
- Evaluate all projects using the standardized 100-mark rubric.
- Publish regional shortlist and presentation instructions.

PHASE 07 | REGIONAL SELECTION

- Conduct regional presentations for shortlisted students.
- Recommended format: 5-minute presentation + 3-minute jury Q&A and it will be extended as per nature of project.
- Evaluate innovation, AI/ML application, Design Thinking, prototype and impact.
- Create the final pool of 30 National Finalists.

PHASE 08 | NATIONAL GRAND FINALE

- Invite 30 National Finalists to the Grand Finale.
- Conduct project exhibition, demonstrations and expert interactions.
- Organize final jury presentations and Q&A.
- Complete final scoring, moderation and jury deliberation.

PHASE 09 | RECOGNITION, IMPACT & LEGACY

- Present national, regional and special recognition awards.
- Issue certificates to participants, finalists, mentors and schools.
- Publish event highlights and winning project stories.
- Create pathways for mentoring, incubation, exhibitions or further development.
- Prepare a post-event impact report for schools, partners and stakeholders.

3. Design Thinking + AI/ML Project Framework

- 1. Empathize** — Understand users, stakeholders and the context of the problem.
- 2. Define** — Convert observations into a clear, focused problem statement.
- 3. Ideate** — Generate multiple possible solutions and select the strongest concept.
- 4. Prototype** — Build an AI/ML-enabled prototype or working model.
- 5. Test** — Validate the solution using data, user feedback or practical testing.
- 6. Improve** — Refine the solution based on evidence and feedback.
- 7. Present** — Demonstrate the solution, technical approach, impact and future scope.

4. Suggested Areas for project

- Healthcare
- Environment & Sustainability
- Agriculture
- Rural Areas & Communities
- Accessibility & Inclusion for persons with disabilities
- Social Impact

5. Evaluation Framework — 100 Marks

Evaluation Parameter	Weightage
Problem Identification & Relevance	10
Design Thinking Process	15
Innovation & Creativity	15
AI / ML Application	20
Prototype / Working Model	15
Practicality & Scalability	10
Social / Environmental Impact	10
Presentation & Q&A	5
TOTAL	100

6. Selection Funnel

800 Schools → 800 Students → 800 Projects → 100 Regional Shortlisted Projects → 30 National Finalists → 1 National Champion

Awards & Recognition

- National AI Innovation Champion 1
- National First Runner-Up 1
- National Second Runner-Up 1
- Best AI/ML Project
- Best Design Thinking Project
- Best Social Impact Project
- Best Prototype / Working Model
- Most Innovative Idea
- Best School Innovation
- Young AI Innovator Award

7. Master Timeline

Date	Milestone	Key Activities	Expected Outcome
02-Sep-26	Competition Launch PAN India	Official launch of the Aryabhata Young Innovators Challenge by SHARD. Nationwide school outreach, promotion and awareness campaign begins.	Schools across India are informed about the competition, eligibility, process and participation opportunities.
02 Sep – 30 Sep 2026	School Registration & Project Nomination	Schools register for the competition and nominate participating students/teams and their project concepts.	Approximately 800 projects registered from schools across different regions of India.
1 Oct – 25 Oct 2026	Project Development & Internal School Competition	Students identify real-world problems, develop AI/ML and innovation-based solutions, build prototypes, test their ideas and present them at the school-level internal competition .	Each participating school selects and submits its best project for the national-level competition.
26 – 27 Oct 2026	Final Submission & Screening Readiness	Submission of final project details, prototype/model evidence, presentation material and required documentation.	All nominated projects are complete and ready for national screening.
28-30 Oct-26	Round 1 – National Screening	SHARD evaluates submitted projects on defined innovation, technical, impact and presentation criteria.	100 projects shortlisted from approximately 800 submissions .
01 Nov – 14 Nov 2026	Mentoring, Refinement & Presentation Preparation	Shortlisted teams receive mentoring and refine their prototype, documentation, solution impact and final presentation.	100 shortlisted teams strengthen their projects for Round 2 evaluation.
15-Nov-26	Round 2 – National Screening	Detailed evaluation of the 100 shortlisted projects based on innovation, problem relevance, technical implementation, AI/ML integration, impact and presentation.	30 National Finalist Projects selected.
16 – 24 Nov 2026	Finalist Preparation & Jury Readiness	Finalists refine their solutions, demonstrations, documentation and presentations for the final evaluation.	30 National Finalists prepared for the final jury round.

25-Nov-26	Round 3 – Final Jury Screening by Framez AI	Final evaluation of the 30 national finalists by the Framez AI, USA jury/body based on the competition's final evaluation criteria.	Top 13 projects selected for the National Grand Finale.
To Be Announced	National Grand Finale	The Top 13 teams present their projects before the final jury and compete for national recognition and awards.	National Winners, Runners-up and Special Awardees announced.

8. Recommended Governance Structure

- National Steering Committee — Overall strategy, governance and approvals.
- National Event Secretariat — Central coordination, communication and reporting.
- Regional Coordinators — School outreach, regional operations and event execution.
- Academic & Technical Panel — Curriculum, project standards and technical guidance.
- Jury & Evaluation Panel — Independent project evaluation and scoring.
- School Coordinators — Student registration, communication and project support.
- Mentors / Experts — Technical, innovation and presentation guidance.

9. Key Success Metrics

- 800 schools on-boarded across four regions.
- 800 student innovators participating.
- 800 AI/ML and Design Thinking projects submitted.
- 100 regional projects shortlisted.
- 30 National Finalists selected.
- 1 National Champion and consolation prizes.
- Strong school, educator, industry and innovation-partner engagement.
- Documented student learning, innovation and social-impact outcomes.

10. Strategic Positioning

The initiative should be positioned as a **Aryabhata Young Innovators Challenge** rather than a conventional coding competition. AI/Machine Learning will form the technology layer, while Design Thinking will provide the human-centered problem-solving framework. This approach allows students from different technical backgrounds to participate while keeping the focus on innovation, relevance, responsible technology and measurable impact.